SupportShift: A Theory-Linked Spatial Simulation Benchmark for Ignored Matérn Observation Support

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ABSTRACT

Spatial measurements often represent pixels, footprints, or local averages but are fitted as point observations. We introduce *SupportShift*, a parameterized and scalable synthetic benchmark that separates the physical covariance parameter, the population Kullback–Leibler (KL) target, and finite-sample estimation error under this support misspecification. Users can vary support width and geometry, smoothness, lattice size, output dimension, replicate count, and covariance-library resolution. Its analytic anchor is a stationary Matérn Gaussian field in $\mathbb{R}^d$ observed through a compact symmetric averaging kernel of bandwidth h. For a fixed nonzero-lag two-point likelihood with known smoothness ν, the naive inverse-range shift has order $h^{2\nu}$ for $0 < \nu < 1$, $h^2 \log(1/h)$ at $\nu = 1$, and h^2 for $\nu > 1$. Explicit coefficients prove range inflation for every fixed $\nu > 0$ at sufficiently small support. A transition-aware two-term approximation resolves the slow finite-bandwidth crossover at $\nu = 1$, while a directional h^2 contrast handles elongated support. For N independent p-dimensional Gaussian fields and a deterministic finite covariance library, a simultaneous likelihood certificate scales as $\sqrt{\log(M)/(Np)} + \log(M)/(Np)$ under relative spectral control; spatial dependence within each field is unrestricted. Four reproducible tracks pair these statements with deterministic quadrature, finite-grid likelihoods, directional support, and growing-p replicated fields. Released generators, source tables, seeds, metadata, and pass/fail gates make every reported number auditable. The benchmark exposes a sharp failure mode: likelihood noise can vanish while a point-support fit concentrates around the wrong range target.

CCS CONCEPTS

• **Computing methodologies → Modeling and simulation**; **Model verification and validation**; • **Mathematics of computing** → *Probabilistic representations.*

KEYWORDS

Matérn covariance, change of support, Gaussian random field, high-dimensional probability, spatial simulation benchmark, misspecified likelihood

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1 INTRODUCTION

Values recorded on pixels, grid cells, or after kernel preprocessing are local aggregates of an underlying spatial field. Change-of-support models account for this by integrating the latent process over the measurement support [3, 8, 16, 18]. A common shortcut instead assigns each aggregate to a declared center and fits a point-support covariance. Relative to the latent parameters, the support is then misinterpreted. A single two-point covariance can still be matched exactly by different variance and decay parameters; a multi-lag likelihood is genuinely misspecified and selects a Kullback–Leibler (KL) projection [1].

This qualitative fact leaves quantitative and experimental questions. How quickly does the pseudo-parameter move as support shrinks? Does the rate depend on Matérn smoothness? Can directional support masquerade as directional range? Finally, does increasing dimension or replication remove the error, or merely make a wrong population target more precise? The first answer is not an ordinary second-order Taylor expansion: the Matérn covariance changes regularity at the origin as smoothness crosses one.

Block covariance and regularized variograms are classical [4, 8, 9]. Classical semivariogram regularization also shows that larger sampling support can increase an apparent or practical range. Clark's geometric rule concerns a finite range of influence, while Bellehumeur and Legendre use model-specific de-regularization approximations [2, 5]. A Matérn correlation has no finite range of influence; our target is instead the variance-refitted KL parameter at a fixed nonzero lag. Fuentes explicitly represents rectangular block averages in the spectral likelihood and invokes a small-pixel approximation [7]; the next-order point-fit parameter displacement is not part of that approximation. Modern aggregated-GP and INLA–SPDE methods likewise put the support operator inside the observation model [16, 18]. Convolution and covariance anisotropy can also be confounded [6, 14]. Generic misspecified-GP theory supplies the KL framework [1], while Gaussian quadratic-form concentration is standard [11, 13]. Table 1 separates these known ingredients from our claims.

We contribute a theorem-linked benchmark rather than a new generic change-of-support method:

- We derive an all-smoothness, d-dimensional fixed-lag phase law with explicit coefficients, a universal small-support range direction, and a two-term approximation continuous through the smoothness-one threshold.
- We derive a directional h^2 apparent-range contrast for elongated support. The qualitative link between convolution and anisotropy is known.
- We certify finite-library likelihood concentration and ERM excess risk for N independent p-dimensional Gaussian fields without assuming independent spatial coordinates.
- SupportShift turns the theorem into four reusable synthetic tracks: a continuous phase oracle, finite-grid likelihoods, directional support, and growing-p replicated fields. Each

exposes declared scale controls, deterministic seeds, source tables, numerical gates, and a machine-checkable claim audit.

The phase theorem is pairwise; full-grid targets are computed, not asserted to share its coefficient. The high-dimensional experiment grows domain at fixed lattice spacing and uses independent replicated fields. We do not claim separate fixed-domain consistency of Matérn variance and range, which would conflict with microergodicity results [12, 17].

2 MODEL AND EXACT PAIR TARGET

Let $Y = \{Y(t) : t \in \mathbb{R}^d\}$ be a mean-zero stationary Gaussian field with

$$C(r) = v\mathcal{M}_\nu(\alpha\|r\|), \qquad \mathcal{M}_\nu(x) = \frac{2^{1-\nu}}{\Gamma(\nu)} x^\nu K_\nu(x), \tag{1}$$

where $v, \alpha, \nu > 0$ and $\mathcal{M}_\nu(0) = 1$. Thus α is inverse range. Let k be a symmetric probability density supported on $B(0, L)$, set $k_h(u) = h^{-d}k(u/h)$, and observe

$$Z_h(t) = \int k_h(u)Y(t-u)\,du. \tag{2}$$

For independent $U, V \sim k$, write $D = U - V$, $\Sigma_k = \mathbb{E}(UU^\top)$, $T_k = \operatorname{tr}(\Sigma_k)$, and $m_q = \mathbb{E}\|D\|^q$. Then $m_2 = 2T_k$, and stationarity gives

$$C_h(r) = v\mathbb{E}[\mathcal{M}_\nu\{\alpha\|r + hD\|\}]\,. \tag{3}$$

Assume $T_k > 0$.

Fix $r = Re$, $R > 0$, and $\|e\| = 1$. A naive point-support pair model with known ν and unknown variance and inverse range uses

$$v^\dagger \begin{pmatrix} 1 & \mathcal{M}_\nu(\alpha^\dagger R) \\ \mathcal{M}_\nu(\alpha^\dagger R) & 1 \end{pmatrix}. \tag{4}$$

The smoothed two-by-two covariance lies in this family. Moreover, $0 < \rho_h(r) < 1$: positivity follows from $k \geq 0$ and $\mathcal{M}_\nu > 0$, while strict inequality follows from the positive smoothed spectral density near the origin. Hence the exact Gaussian KL target is

$$v_h^\dagger = C_h(0), \qquad \mathcal{M}_\nu(\alpha_h^\dagger R) = \rho_h(r) := \frac{C_h(r)}{C_h(0)}. \tag{5}$$

It is unique because

$$\mathcal{M}_\nu'(x) = -\frac{2^{1-\nu}}{\Gamma(\nu)} x^\nu K_{\nu-1}(x) < 0. \tag{6}$$

A standard route to pair-composite consistency would use a fixed lattice, increasing rectangular Følner domains, a compact parameter space, target uniqueness and interiority, correlations bounded away from one, and a uniform ergodic law with an integrable envelope. We do not establish that uniform law. Dependence among overlapping pairs would require Godambe uncertainty, not an independence Hessian.

3 SMOOTHNESS-DEPENDENT PHASE LAW

Put $z = \alpha R$, $a_k(e) = e^\top \Sigma_k e$, and

$$B_{\nu,k}(r) = \alpha^2 \left[a_k(e)\mathcal{M}_\nu''(z) + \{T_k - a_k(e)\}\frac{\mathcal{M}_\nu'(z)}{z} \right]. \tag{7}$$

Proposition 1 (Fixed nonzero lag). *Uniformly for r in a compact annulus bounded away from zero, for $h \leq \|r\|/(4L)$, and for α in a compact subset of $(0, \infty)$,*

$$\frac{C_h(r)}{v} = \mathcal{M}_\nu(\alpha\|r\|) + h^2 B_{\nu,k}(r) + O(h^4). \tag{8}$$

At zero lag, the Matérn origin expansion produces a phase transition.

Theorem 1 (Matérn support phase law). *Fix $\nu > 0$. Let*

$$c_\nu = \frac{\Gamma(1-\nu)}{\nu 2^{2\nu}\Gamma(\nu)} > 0, \qquad 0 < \nu < 1.$$

Under the preceding assumptions, as $h \downarrow 0$,

$$\frac{C_h(0)}{v} = 1 - c_\nu m_{2\nu}(\alpha h)^{2\nu} + O(h^2), \qquad 0 < \nu < 1, \tag{9}$$

$$= 1 - \frac{m_2}{2}(\alpha h)^2 \log\{1/(\alpha h)\} + O(h^2), \qquad \nu = 1, \tag{10}$$

$$= 1 - \frac{m_2}{4(\nu-1)}(\alpha h)^2 + o(h^2), \qquad \nu > 1. \tag{11}$$

The pseudo-decay in (5) *satisfies*

$$\alpha_h^\dagger - \alpha = \frac{\mathcal{M}_\nu(z)c_\nu m_{2\nu}\alpha^{2\nu}}{R\mathcal{M}_\nu'(z)} h^{2\nu} + O\left(h^{\min(4\nu,2)}\right), \quad 0 < \nu < 1, \tag{12}$$

$$\alpha_h^\dagger - \alpha \sim \frac{m_2\mathcal{M}_1(z)\alpha^2}{2R\mathcal{M}_1'(z)} h^2 \log(1/h), \qquad \nu = 1, \tag{13}$$

$$\alpha_h^\dagger - \alpha = \frac{A_{\nu,k}(r)}{R\mathcal{M}_\nu'(z)} h^2 + \mathcal{R}_\nu(h), \qquad \nu > 1, \tag{14}$$

where

$$A_{\nu,k}(r) = \alpha^2\{(2\nu-2)a_k(e) + T_k\}G_\nu(z) > 0, \tag{15}$$

$$G_\nu(z) = \frac{\mathcal{M}_\nu(z)}{2\nu-2} + \frac{\mathcal{M}_\nu'(z)}{z} \tag{16}$$

$$= \frac{2^{1-\nu}}{\Gamma(\nu)} \frac{z^\nu K_{\nu-2}(z)}{2\nu-2} > 0, \tag{17}$$

and $\mathcal{R}_\nu(h) = O(h^{2\nu})$ for $1 < \nu < 2$, $O\{h^4\log(1/h)\}$ for $\nu = 2$, and $O(h^4)$ for $\nu > 2$. For fixed ν, the remainders are uniform over compact positive decay sets and lag annuli bounded away from zero, but not as ν approaches one or two. Consequently, for every $\nu > 0$ and all sufficiently small $h > 0$,

$$v_h^\dagger < v, \qquad \rho_h(r) > \mathcal{M}_\nu(\alpha R), \qquad \alpha_h^\dagger < \alpha.$$

The leading law is pointwise in fixed ν. To resolve its slow crossover, let $b_\nu = \Gamma(-\nu)/\{2^{2\nu}\Gamma(\nu)\}$ and define, for $0 < \nu < 2$,

$$W_{\nu,k}(h) = \begin{cases} \dfrac{m_2(\alpha h)^2}{4(1-\nu)} + b_\nu m_{2\nu}(\alpha h)^{2\nu}, & \nu \neq 1, \\ \dfrac{(\alpha h)^2}{2}\left[m_2\{\log(\alpha h/2) + \gamma_{\mathrm{E}} - 1/2\} + \ell_{2,k}\right], & \nu = 1, \end{cases} \tag{18}$$

where $\ell_{2,k} = \mathbb{E}\{\|D\|^2 \log\|D\|\}$. Set

$$\widetilde{\rho}_h = \frac{\mathcal{M}_\nu(z) + h^2 B_{\nu,k}(r)}{1 + W_{\nu,k}(h)}, \qquad \mathcal{M}_\nu(\widetilde{\alpha}_h R) = \widetilde{\rho}_h. \tag{19}$$

Table 1: Nearest known ingredients and the narrower SupportShift addition.

Literature	Established result	Addition studied here
Change of support and block covariance [4, 7, 8, 16]	Integrate the latent covariance over known observation supports	Explicit small-support Matérn *point-fit pseudo-range* phase law
Regularized-semivariogram range rules [2, 5]	Support can enlarge a finite or practical variogram range	Infinite-range Matérn KL target, smoothness phase, and coefficient
Misspecified GP likelihood [1]	Population KL pseudo-target under covariance misspecification	Closed direction, rate, and coefficient for ignored support
Convolution and anisotropy [6, 14]	Support/convolution can create anisotropic covariance	Leading directional apparent-range contrast
Gaussian quadratic forms [11, 13]	Nonasymptotic concentration of Gaussian quadratic forms	Direct certificate for the benchmark's structured covariance library
Spatial benchmarks [10]	Shared synthetic data and auditable method comparisons	Support-specific generator with analytic targets and pass/fail gates

Proposition 2 (Transition-aware approximation). *For fixed* $0 < \nu < 2$,

$$\alpha_h^{\dagger} - \widetilde{\alpha}_h = \begin{cases} O(h^{2\nu+2}), & 0 < \nu < 1, \\ O\{h^4 \log(1/h)\}, & \nu = 1, \\ O(h^4), & 1 < \nu < 2. \end{cases} \tag{20}$$

Moreover $W_{\nu,k}(h) \to W_{1,k}(h)$ *as* $\nu \to 1$ *for fixed* $h > 0$. *The error orders are not claimed jointly uniform in* (ν, h).

Proposition 3 (Directional support contrast). *Let* $\Delta_e(h) = \alpha - \alpha_h^{\dagger}(e)$. *For two unit lag directions* e_1, e_2, *under the theorem assumptions,*

$$\Delta_{e_1}(h) - \Delta_{e_2}(h) = \frac{\alpha^2\{a_k(e_1) - a_k(e_2)\}}{R} \frac{K_{\nu-2}(z)}{K_{\nu-1}(z)} h^2 + o(h^2). \tag{21}$$

The formula holds for every $\nu > 0$. *When* $\nu \le 1$, *it is lower order than the direction-independent leading shift in the phase theorem.*

For the benchmark, let

$$Z_{h,A}(t) = \int k(u) Y(t - hAu)\, du.$$

Then $\Sigma_k = AA^\top/5$. We vary the principal singular-value aspect ratio $\varrho \ge 1$ while holding $\operatorname{tr}(\Sigma_k)$ fixed using

$$A_\varrho = \sqrt{\frac{2}{\varrho + \varrho^{-1}}}\, \operatorname{diag}(\sqrt{\varrho}, \varrho^{-1/2}). \tag{22}$$

For $\nu > 1$, the ratio of the major- to minor-axis leading shift coefficients is

$$\frac{(2\nu - 1)\varrho^2 + 1}{\varrho^2 + 2\nu - 1}. \tag{23}$$

Thus elongated support can induce direction-dependent inferred range even for an isotropic latent covariance.

Remark 1. The fitted lag R is fixed and bounded away from zero. If $R = ch$, the nonzero-lag expansion is no longer uniform and the relative displacement need not vanish.

Thus ignored support inflates inferred range. The sign is not an assumption: for $\nu > 1$, it follows from positivity of $K_{\nu-2}$ in the displayed identity. For an isotropic kernel, $A_{\nu,k}(r) = \sigma_k^2 \alpha^2 (2\nu + d - 2) G_\nu(z)$.

3.1 Proof sketch

Let $f_\alpha(x) = \mathcal{M}_\nu(\alpha\|x\|)$. At a nonzero lag, fourth-order Taylor expansion is uniform. Symmetry and $\mathbb{E}(DD^\top) = 2\Sigma_k$ cancel odd terms, including the cubic term, and yield (8). At zero, the noninteger Bessel expansion is

$$\mathcal{M}_\nu(x) = \sum_{j\ge0} \frac{(x^2/4)^j}{j!(1-\nu)_j} + \frac{\Gamma(-\nu)x^{2\nu}}{2^{2\nu}\Gamma(\nu)} \sum_{j\ge0} \frac{(x^2/4)^j}{j!(1+\nu)_j}. \tag{24}$$

Taking expectations at $x = \alpha h\|D\|$ proves (9) and (11). At $\nu = 1$,

$$\mathcal{M}_1(x) = 1 + \frac{x^2}{2}\{\log(x/2) + \gamma_{\mathrm{E}} - 1/2\} + O\{x^4(1 + |\log x|)\},$$

which proves (10). Divide (8) by the relevant variance expansion and expand the inverse of $\alpha \mapsto \mathcal{M}_\nu(\alpha R)$. This gives (12)–(14). For $\nu > 1$, the Matérn differential equation and the recurrence

$$K_\nu(z) = K_{\nu-2}(z) + 2(\nu - 1)K_{\nu-1}(z)/z$$

reduce the normalized-correlation coefficient to $A_{\nu,k}(r)$, proving its sign. Appendix A records the proof details used by these claims; the reproducible technical manuscript also gives an exact exponential benchmark.

For (21), subtract the two directional fixed-lag expansions; the common zero-lag normalization and all direction-free terms cancel. Using

$$\mathcal{M}_\nu''(z) - \mathcal{M}_\nu'(z)/z = \frac{2^{1-\nu}}{\Gamma(\nu)} z^\nu K_{\nu-2}(z)$$

and dividing by $-R\mathcal{M}_\nu'(z)$ gives the displayed coefficient.

At the remaining integer transition,

$$\mathcal{M}_2(x) = 1 - \frac{x^2}{4} + x^4\left\{\frac{3}{64} - \frac{\log(x/2) + \gamma_{\mathrm{E}}}{16}\right\} + O\{x^6(1 + |\log x|)\}.$$

For fixed $\nu > 2$, four-times differentiability yields $\mathcal{M}_\nu(x) = 1 - x^2/\{4(\nu - 1)\} + O(x^4)$. These formulas, and the fractional term for $1 < \nu < 2$, give the three stated smooth-regime remainders after division and inverse expansion.

4 FINITE-GRID SUPPORT-AWARE LIKELIHOOD

The theorem covers translation-invariant interior averaging. For a finite simulation, let X_{in} denote raw sites and let $S_h \in \mathbb{R}^{m\times n}$ be a rectangular row-stochastic observation operator. With noise present before smoothing,

$$y = S_h\{Y(X_{\text{in}}) + \epsilon\}, \qquad \epsilon \sim N(0, \tau^2 I_n), \tag{25}$$

the support-aware covariance is

$$\Sigma_{\text{corr}}(\alpha) = S_h\{K_\alpha(X_{\text{in}}, X_{\text{in}}) + \tau^2 I_n\}S_h^\top. \tag{26}$$

The naive declared-center point-support covariance is instead

$$\Sigma_{\text{naive}}(\alpha) = K_\alpha(X_{\text{out}}, X_{\text{out}}) + \tau^2 I_m. \tag{27}$$

Besides ignoring support, (27) treats pre-smoothing noise as independent post-smoothing noise.

With variance, smoothness, and nugget fixed, the finite-design KL target minimizes

$$Q_n(\alpha) = \frac{1}{2}\left[\log\det\Sigma(\alpha) + \operatorname{tr}\{\Sigma(\alpha)^{-1}\Sigma_0\}\right]. \tag{28}$$

The corrected family contains Σ_0 when the operator and fixed nuisance parameters are correctly specified and decay is identifiable; the naive family generally does not. Our primary experiment sets $\tau = 0$, isolating field support from noise-timing misspecification. A separate $\tau^2 = 0.05$ run is retained only as a combined-misspecification artifact. Rectangular S_h is intentional: an invertible square, parameter-independent transform may merely change coordinates, whereas aggregation reduces raw sites to fewer supported outputs.

Boundary row normalization is outside the interior theorem. If two location-dependent scaled kernels have means μ_s and μ_t, then

$$C_{h,s,t}(r) = C(r) + h\nabla C(r)^\top(\mu_s - \mu_t) + O(h^2).$$

A truncated kernel can therefore create a first-order term and destroy stationarity; the interior theorem gives no boundary sign guarantee. We include one boundary and one fixed irregular design as illustrations, not as a boundary scaling study.

4.1 Finite-library high-dimensional certificate

The growing-p benchmark jointly varies variance and decay on a deterministic finite grid. The following standard quadratic-form consequence explains which part of its error can shrink with information; it is supporting machinery, not a new concentration inequality.

PROPOSITION 4 (GAUSSIAN FINITE-LIBRARY CERTIFICATE). *Let $0 < \delta < 1, \Sigma_0 \in \mathbb{S}^p_{++}$, and $X_1, \ldots, X_N \overset{\text{iid}}{\sim} N_p(0, \Sigma_0)$. Let $\{\Sigma_\theta : \theta \in \mathcal{G}\}$ be a deterministic library of M covariances in $\mathbb{S}^p_{++}$. Define*

$$\widehat{\Sigma}_N = N^{-1}\sum_{i=1}^N X_i X_i^\top, \tag{29}$$

$$\widehat{L}_N(\theta) = \frac{1}{2p}\left\{\log\det\Sigma_\theta + \operatorname{tr}(\Sigma_\theta^{-1}\widehat{\Sigma}_N)\right\}, \tag{30}$$

$$L(\theta) = \mathbb{E}\widehat{L}_N(\theta), \qquad A_\theta = \Sigma_0^{1/2}\Sigma_\theta^{-1}\Sigma_0^{1/2}. \tag{31}$$

For $t = \log(2M/\delta)$, with probability at least $1-\delta$, simultaneously for every $\theta \in \mathcal{G}$,

$$|\widehat{L}_N(\theta) - L(\theta)| \le \frac{\|A_\theta\|_F}{p}\sqrt{\frac{t}{N}} + \frac{\|A_\theta\|_{\text{op}}}{p}\frac{t}{N} =: u_\theta. \tag{32}$$

If $\widehat{\theta}$ and $\theta^\star$ minimize the empirical and population criteria, respectively, then

$$L(\widehat{\theta}) - L(\theta^\star) \le 2\max_{\theta\in\mathcal{G}} u_\theta. \tag{33}$$

Moreover, $L(\theta)$ differs by a candidate-independent constant from

$$p^{-1}D_{\text{KL}}\{N(0,\Sigma_0)\,\|\,N(0,\Sigma_\theta)\}. \tag{34}$$

To prove (32), write $X_i = \Sigma_0^{1/2}Z_i$, stack the standard Gaussian vectors, and apply the Gaussian quadratic-form inequality to $I_N \otimes A_\theta$. Its Frobenius and operator norms are $\sqrt{N}\|A_\theta\|_F$ and $\|A_\theta\|_{\text{op}}$; division by $2Np$ and a union bound give the displayed constants [11, 13]. The ERM result is the usual three-term comparison through $\widehat{L}_N$.

No coordinate independence is used: all within-field spatial dependence is in Σ_0. Writing

$$r_2(A_\theta) = \frac{\|A_\theta\|_F^2}{\|A_\theta\|_{\text{op}}^2},$$

the radius in (32) is

$$\frac{\|A_\theta\|_{\text{op}}}{p}\left\{\sqrt{\frac{r_2(A_\theta)t}{N}} + \frac{t}{N}\right\}.$$

Thus the relevant dimension is spectral: increasing p helps only when the operator norm and effective-rank geometry remain controlled. If $c\Sigma_0 \preceq \Sigma_\theta \preceq C\Sigma_0$ uniformly, then

$$\max_\theta u_\theta \le c^{-1}\left\{\sqrt{\frac{\log(2M/\delta)}{Np}} + \frac{\log(2M/\delta)}{Np}\right\}. \tag{35}$$

This conditional rate does not imply separate fixed-domain range and variance consistency: relative spectra or parameter curvature may deteriorate with p. The benchmark therefore reports the actual matrix norms and the exact candidate-specific radius.

5 SYNTHETIC VALIDATION

The public benchmark contract has four components in every track: declared generating factors, a nonrandom oracle or population target, stochastic or quadrature output, and a numerical pass/fail gate. A method can therefore be evaluated against the physical parameter and its own support-specific KL target rather than against one ambiguous "truth." The generator exposes bandwidth, smoothness, support geometry, lattice size, replication, and candidate-grid resolution; saved metadata records their resolved values, seeds, environment, and hashes. The release verifier rejects incomplete factor grids, failed gates, or altered paper inputs. This contract makes the data reusable for alternative estimators while keeping comparisons inside the declared model and target.

5.1 Continuous phase oracle

We used a two-dimensional product Epanechnikov density, $\alpha = R = 1$, tensor Gauss–Legendre order 96, 18 bandwidths from 0.003 to 0.3, and $\nu \in \{0.25, 0.5, 0.75, 1, 1.5, 2.5\}$. The density of each coordinate of D is available in closed form, so (3) is evaluated by deterministic quadrature. We solved (5) by monotone root finding. For this kernel $L = \sqrt{2}$, so the two largest displayed bandwidths, 0.229 and 0.3, are stress points outside the sufficient Taylor neighborhood $h \le R/(4L)$. All slope and coefficient checks use the six smallest bandwidths, which are inside that neighborhood. Repeating at orders 64 and 128 changed pseudo-decay by at most 4.4×10^{-8} and 5.2×10^{-9}, respectively, relative to order 96.

All 108 shifts $\alpha - \alpha_h^{\dagger}$ were positive. For smoothness 0.25, 0.5, and 0.75, log–log slopes over the six smallest bandwidths were 0.515, 0.999, and 1.468, versus theoretical powers 0.5, 1, and 1.5. The raw slope at $\nu = 1$ was 1.817, reflecting the logarithmic modifier. Slopes for $\nu = 1.5$ and 2.5 were 1.994 and 2.000. At the smallest bandwidth, ratios of the exact shift to the full theorem leading term were 1.021, 1.000, 0.958, 1.059, 0.997, and 1.000 in increasing order of ν.

The pointwise qualifier matters at practical bandwidths. We therefore audited the threshold layer on 111 deterministic cells with $\nu \in [0.55, 1.45]$, spacing 0.025, and $h \in \{0.01, 0.02, 0.05\}$. The smallest exact-shift to one-term ratio was 0.154; equivalently, the one-term prediction was up to 6.51 times the exact shift near, but not at, $\nu = 1$. In contrast, the maximum relative error of $\widetilde{\alpha}_h$ was 0.099%, and the maximum relative error in variance loss was 0.084%. Every exact and approximate shift had the theorem-predicted sign. Repeating the exact target at quadrature orders 64 and 128 changed it by at most 4.0×10^{-9} and 3.1×10^{-10}, respectively, relative to the shift computed at order 96. These are finite-grid diagnostics, not a claimed joint uniform asymptotic theorem.

5.2 Directional support oracle

The directional track uses (22) with $\varrho \in \{1, 4\}$, 19 lag angles from zero to 90 degrees, $\nu \in \{0.5, 1, 1.5, 2.5\}$, and 14 bandwidths from 0.003 to 0.15. All 2,128 inverse-range shifts were positive. At $\varrho = 4$, the major/minor smooth-regime coefficient ratios were 1.833 for $\nu = 1.5$ and 3.250 for $\nu = 2.5$, exactly as predicted by (23). At $h = 0.15$, the inferred range inflation along the major and minor axes was 13.46% and 11.47% for $\nu = 0.5$, and 1.56% and 0.76% for $\nu = 1.5$.

The major-minus-minor shift divided by its $D_\nu h^2$ term differed from one by at most 4.3×10^{-6} at the smallest bandwidth over all four smoothnesses. Relative to order 96, orders 64 and 128 changed any individual shift by at most 1.3×10^{-8} of that shift and changed a directional contrast by at most 1.2×10^{-9} of that contrast.

5.3 Finite-grid full likelihood

The core design uses a 19×19 input lattice with spacing 0.25 and every second site as a candidate output center, trimmed by 0.75 in each coordinate. We set $v = \alpha = 1$, $\tau^2 = 0$, $\nu \in \{0.5, 1, 1.5, 2.5\}$, and $h \in \{0, 0.3, 0.5, 0.7\}$. Radial Epanechnikov rows are normalized exactly. Stress designs retain boundary centers, perturb each coordinate uniformly in $[-0.04, 0.04]$ before clipping, and grow one-dimensional domains from 100 to 400 raw sites. The irregular panel is one fixed realization.

Table 2: Finite-design results at $h = 0.7$. MCSE is for the reported mean.

ν	Model	Target	Mean	MCSE	Bound hits
0.5	Corrected	1.000	1.090	0.047	3
0.5	Naive	0.152	0.151	0.004	0
1	Corrected	1.000	0.990	0.010	0
1	Naive	0.414	0.407	0.005	0
1.5	Corrected	1.000	0.990	0.007	0
1.5	Naive	0.573	0.570	0.005	0
2.5	Corrected	1.000	0.994	0.004	0
2.5	Naive	0.740	0.738	0.003	0

For each configuration, (28) was minimized by bounded scalar optimization over $\log \alpha \in [\log(0.02), \log(5)]$. Two hundred replicate likelihoods were evaluated on a 161-point log-decay profile; interior minima were refined by three-point quadratic interpolation and endpoint minima were retained as boundary fits. Gaussian raw-observation vectors were transformed by S_h. Common random numbers were reused across bandwidths. No adaptive diagonal jitter was added; a Cholesky or population-optimization failure would abort the configuration and make the reducer mark the run incomplete.

The immutable run contained 21 configurations and 8,400 fits. The reducer verified configuration hashes, row counts, and unique model–replicate keys; independent checks covered finiteness, model labels, replicate indices, and metadata. Across the core designs, corrected population targets were within 1.3×10^{-7} of one. At $h = 0.7$, naive targets were 0.152, 0.414, 0.573, and 0.740 as ν increased; Monte Carlo means were 0.151, 0.407, 0.570, and 0.738. Corrected means were 1.090, 0.990, 0.990, and 0.994. The first was within two Monte Carlo standard errors of its target. All six core boundary-hitting fits were retained.

Including boundary outputs changed both output dimension and the rough-field naive target, from 0.152 to 0.135 at $h = 0.7$; for the fixed jittered design the target moved from 0.289 to 0.284 at $h = 0.5$. Corrected targets remained one. In the domain-growth check, increasing raw sites from 100 to 400 reduced corrected-estimator standard deviation from 0.529 to 0.177 and naive-estimator standard deviation from 0.151 to 0.068, while at $n = 400$ their means were within 0.026 and 0.009 of their respective targets.

5.4 Replicated-field high-dimensional track

For $q \in \{4, 6, 8, 10\}$, a padded latent lattice with spacing 0.25 was mapped by a full-row-rank radial Epanechnikov operator with $h = 0.5$ to a $q \times q$ output lattice, giving $p \in \{16, 36, 64, 100\}$. We used $\nu \in \{0.5, 1.5\}$, $v = \alpha = 1$, and $N \in \{1, 4, 16, 64\}$ independent fields. Corrected candidates were $vS_h R_\alpha S_h^\top$; naive candidates were vR_α at output centers. The deterministic $101 \times 161 = 16{,}261$ variance–decay library spans $[0.35, 2.5] \times [0.15, 1.6]$ and contains $(1, 1)$. This truth anchoring is an oracle benchmark control, not a deployable grid-selection rule; the complete library was fixed before the final Monte Carlo run.

Each (p, ν) cell used 200 nested replicate-field sequences, giving 64 model–design–sample-size cells and 12,800 exact finite-library

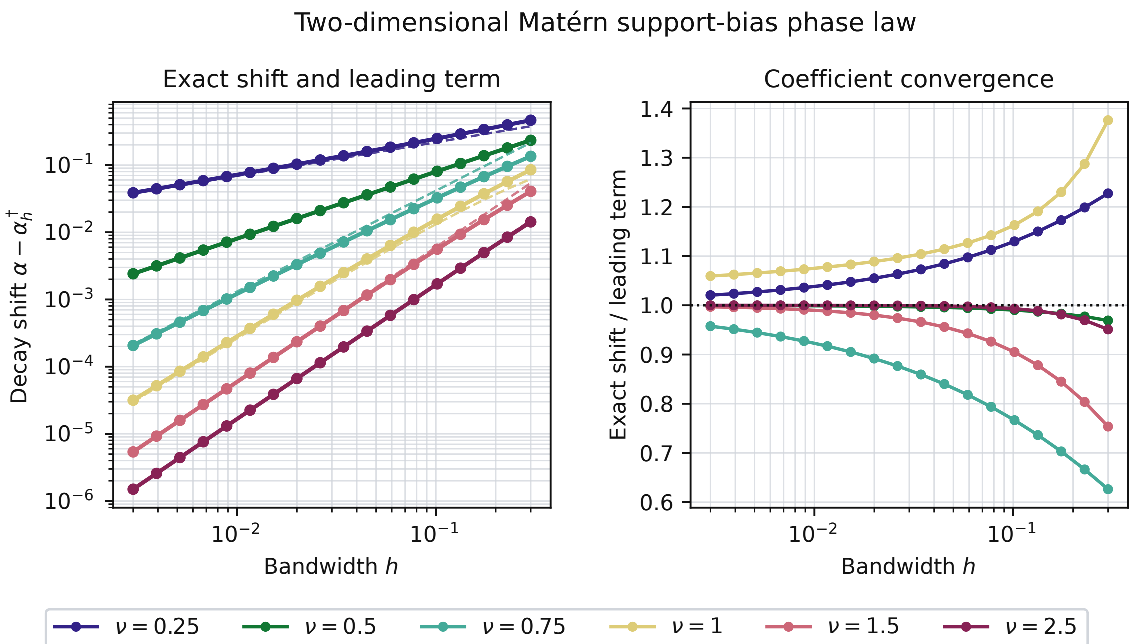

Figure 1: Continuous two-dimensional pseudo-decay shift. Left: exact shifts (solid) and theorem leading terms with explicit coefficients (dashed). Right: their ratio, which approaches one as $h \downarrow 0$.

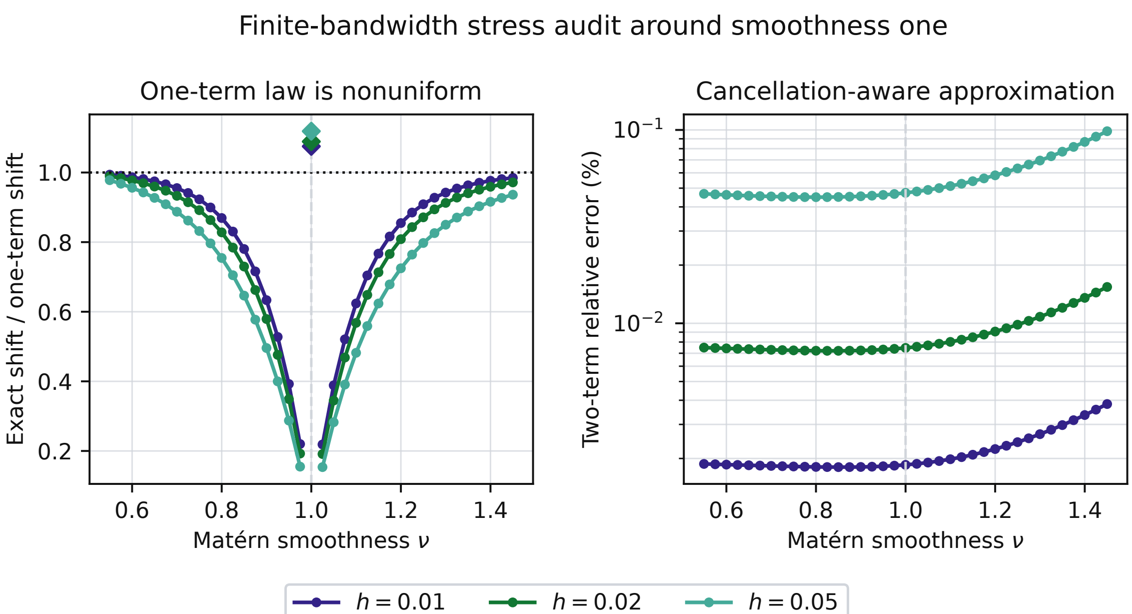

Figure 2: Finite-bandwidth transition stress around $\nu = 1$. Left: the fixed-ν one-term law is accurate away from the crossover but nonuniform near it. Right: retaining the canceling quadratic and fractional terms gives a continuous, sub-0.1% approximation on the audited grid.

fits. The discrete population objective was within 2.63×10^{-5} per coordinate of the continuously profiled objective in all 16 design–model cases, below the 5×10^{-5} tolerance fixed before simulation.

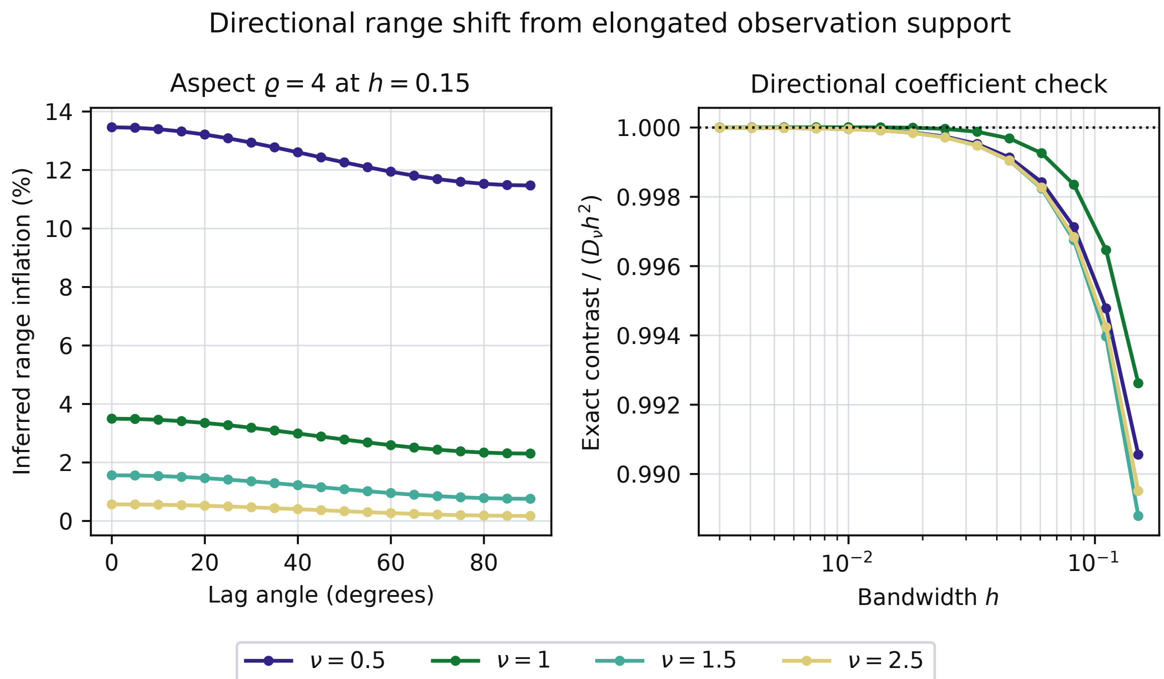

Figure 3: An isotropic latent field observed through elongated support. Left: apparent range inflation versus lag angle at support aspect four and $h = 0.15$. Right: exact major–minor inverse-range contrast divided by $D_\nu h^2$.

No adaptive jitter or coordinate-independence approximation was used. Figure 5 shows the exact observation map for the largest design.

With $\delta = 0.05$, every trial in every cell satisfied the actual candidatewise simultaneous event (32). Across all fits,

$$\max_{\theta} \frac{|\widehat{L}_N(\theta) - L(\theta)|}{u_\theta} \leq 0.789.$$

Each certificate applies separately to its fixed cell, not jointly across all 64. The deterministic ERM inequality (33) also held for every fit. Across cells, the 95th percentile of the worst criterion deviation was at most 0.402 times the worst-candidate radius, with median ratio 0.298. For each fixed model, p, and ν, its log–log slope against N ranged from -0.528 to -0.454, with median -0.505, matching the leading $N^{-1/2}$ dependence.

At $p = 100$, $\nu = 1.5$, the naive finite-grid decay target was 0.594 (continuous target 0.589), versus the physical value one. From $N = 1$ to 64, naive RMSE to its own target fell from 0.124 to 0.015, while its RMSE to the physical value was 0.409 at $N = 64$; the support-aware RMSE fell from 0.206 to 0.024. At $\nu = 0.5, N = 64$, naive RMSE was 0.015 to its target but 0.681 to the physical value. Thus increasing information reduced sampling error without repairing support misspecification. These parameter summaries are empirical; the proposition controls criterion error and excess risk unless a curvature condition is added.

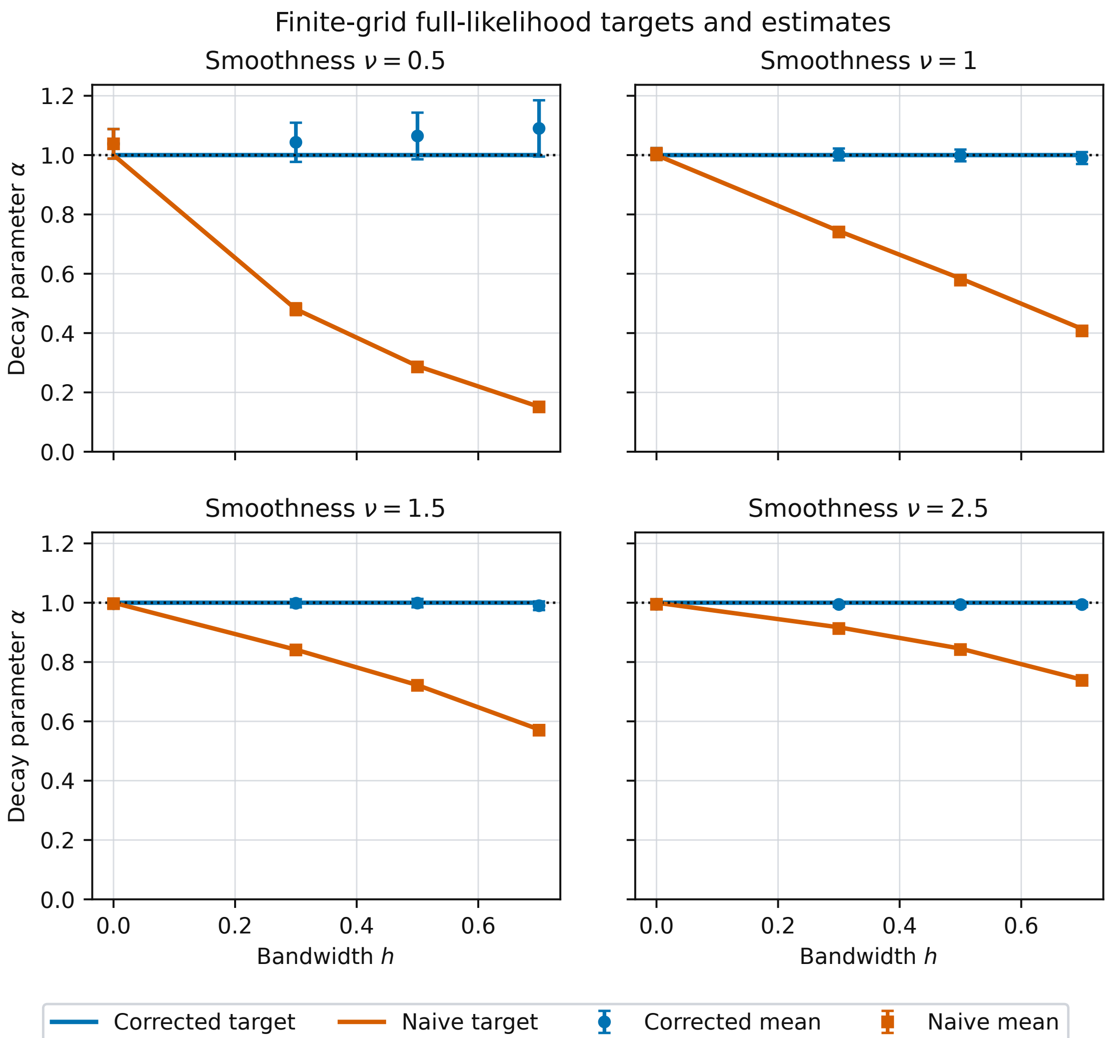

Figure 4: Finite-design full-likelihood targets and Monte Carlo mean estimates. Error bars show two Monte Carlo standard errors of the mean.

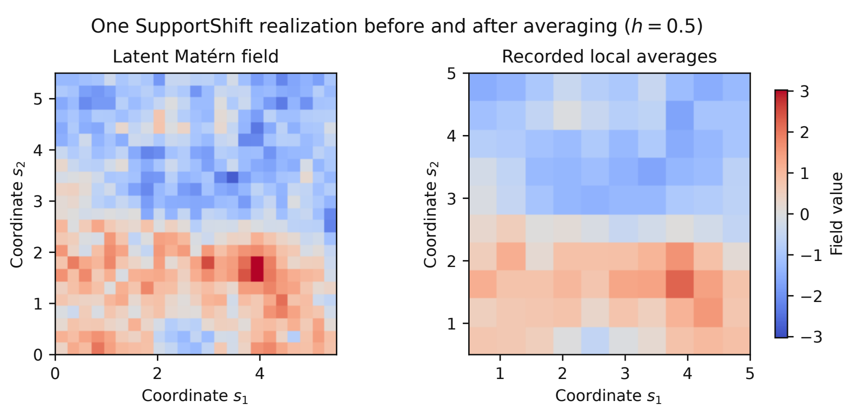

Figure 5: One synthetic field before and after the benchmark's known rectangular support operator.

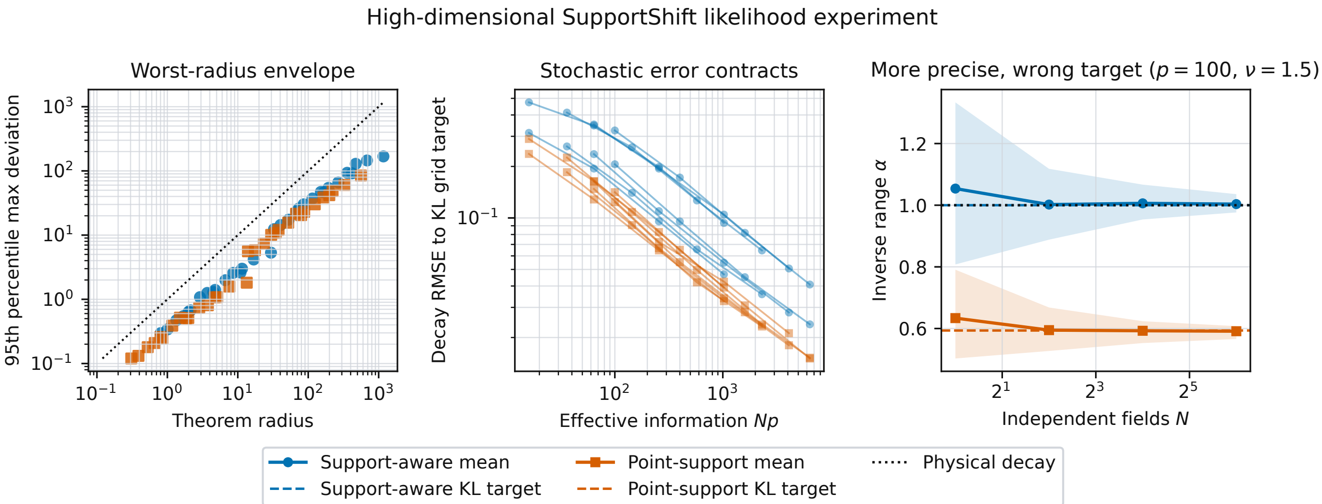

Figure 6: High-dimensional SupportShift likelihood benchmark. Left: the weaker worst-radius envelope; exact candidatewise coverage is reported in text. Center: decay RMSE to each model's finite-grid KL target versus Np. Right: increasing N makes both estimators more precise, but the point-support fit concentrates near its displaced target.

6 DISCUSSION

The theorem separates a standard operation from a less-studied inferential consequence. Propagating a covariance through a known support is classical; the object studied here is the parameter selected when that propagation is ignored. This differs from classical range-of-influence rules: compactly supported variograms can gain a geometric support-width extension [5], but Matérn has no finite range and our variance-refitted KL target moves at a smoothness-dependent power. Rough Matérn fields move at a fractional bandwidth power because the origin cusp changes the variance more rapidly than fixed-lag covariance. Smooth fields move quadratically, but a Bessel identity still fixes the sign.

The simulation has a deliberate role: it validates a theorem-linked generator, computes nonrandom finite-design KL targets, and distinguishes misspecification from finite-sample error. It is not a substitute for the proof. The zero-support case verifies exact agreement, the corrected target verifies model containment, and the boundary and irregular panels expose where the interior theorem stops. The replicated-field track checks the candidatewise probability event and shows that stochastic concentration does not erase KL approximation error.

The scope is narrow. Smoothness and support are known. The original finite-grid track fixes nuisance parameters; the replicated track jointly selects variance and decay only over a declared finite library. The closed target is for a two-point criterion, while full-grid targets are numerical. Fixed-resolution discrete smoothing does not converge to continuous convolution merely because the domain grows. Boundary normalization can introduce first-order terms. Joint support, smoothness, variance, and range estimation may be weakly identified. The exact probability radius uses the generating Σ_0, so it is an oracle benchmark certificate rather than a plug-in interval for an unknown real-data covariance. These are future problems, not implicit claims.

Recent finite-grid Matérn work studies discretization and edge effects without deriving this preprocessing-support pseudo-target phase law [15]. Our narrower contribution is the explicit all-ν expansion, its directional coefficient, and their synthesis with a parameterized theorem-linked benchmark.

The closest analytic setup located in our search is Fuentes's block-average spectral representation and small-pixel approximation [7]. The distinction is that our estimand is the next-order point-support KL parameter after the support transform is omitted. This comparison is a scoped literature conclusion, not a claim of exhaustive priority.

7 CONCLUSION

For a fixed nonzero-lag pair fit under translation-invariant interior averaging, ignored support makes a Matérn field look more persistent than it is. The inverse-range displacement is $h^{2\nu}$, $h^2 \log(1/h)$, or h^2, and its leading sign is negative across fixed smoothness values, dimensions, and compact symmetric kernels. A support-aware covariance removes the finite-design population shift when the known operator and fixed nuisance parameters are correctly specified and decay is identifiable. The result supplies a theorem-level diagnostic and a reproducible simulation protocol. For replicated fields, high-dimensional probability explains how likelihood noise can contract around either the physical covariance or a misspecified KL target. The paper does not assert a universal sign for boundary supports, arbitrary full-likelihood fits, or continuous-parameter estimation.

Code, manifests, and audited aggregate results are available at https://github.com/Sasha-Cui/HighDimSpatialStatistics under the immutable tag `supportshift-geosim-v1.1.2`. The release includes a synthetic-data card, scoped reuse terms, a citation file, and a one-command verifier for the promoted inputs, all paper artifacts, and 100 reported numerical claims.

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A PROOF DETAILS FOR THE ANALYTIC CLAIMS

Proof of Theorem 1. Let $f_\alpha(x) = \mathcal{M}_\nu(\alpha\|x\|)$. Its first four derivatives are uniformly bounded on the declared annulus. Since $D = U - V$ is centrally symmetric and $\mathbb{E}(DD^\top) = 2\Sigma_k$, Taylor's theorem gives

$$\mathbb{E}f_\alpha(r + hD) = f_\alpha(r) + \frac{h^2}{2}\operatorname{tr}\{\nabla^2 f_\alpha(r)\mathbb{E}(DD^\top)\} + O(h^4). \tag{36}$$

For $r = Re$, the radial Hessian is

$$\nabla^2 f_\alpha(r) = \alpha^2\mathcal{M}_\nu''(z)ee^\top + \alpha^2\frac{\mathcal{M}_\nu'(z)}{z}(I - ee^\top),$$

which proves Proposition 1.

For noninteger ν, the convergent origin expansion is

$$\mathcal{M}_\nu(x) = \sum_{j\ge 0}\frac{(x^2/4)^j}{j!(1-\nu)_j} + b_\nu x^{2\nu}\sum_{j\ge 0}\frac{(x^2/4)^j}{j!(1+\nu)_j}, \quad b_\nu = \frac{\Gamma(-\nu)}{2^{2\nu}\Gamma(\nu)}. \tag{37}$$

Because D is bounded, termwise expectation at $x = \alpha h\|D\|$ is dominated uniformly for α in a compact subset of $(0,\infty)$. For $0 < \nu < 1$, the fractional term is leading and the next terms are $O(h^2)$. For $1 < \nu < 2$, the quadratic term is leading and the fractional remainder is $O(h^{2\nu})$. For fixed $\nu > 2$, four-times differentiability at zero gives an $O(h^4)$ remainder. At the two integer transitions,

$$\mathcal{M}_1(x) = 1 + \frac{x^2}{2}\{\log(x/2) + \gamma_{\mathrm{E}} - 1/2\} + O\{x^4(1 + |\log x|)\}, \tag{38}$$

$$\mathcal{M}_2(x) = 1 - \frac{x^2}{4} + x^4\left\{\frac{3}{64} - \frac{\log(x/2) + \gamma_{\mathrm{E}}}{16}\right\} + O\{x^6(1 + |\log x|)\}. \tag{39}$$

These formulas prove (9)–(11), including the stated remainders.

Write $d_h = C_h(0)/v$. Dividing (36) by d_h, then applying the local inverse of $\alpha \mapsto \mathcal{M}_\nu(\alpha R)$, is valid because

$$\mathcal{M}_\nu'(z) = -\frac{2^{1-\nu}}{\Gamma(\nu)}z^\nu K_{\nu-1}(z) < 0.$$

This gives (12)–(14). In the smooth regime, the order-h^2 normalized-correlation coefficient is

$$B_{\nu,k}(r) + \mathcal{M}_\nu(z)\frac{\alpha^2 m_2}{4(\nu-1)}.$$

Using $m_2 = 2T_k$, the Matérn differential equation, and $K_\nu(z) = K_{\nu-2}(z) + 2(\nu-1)K_{\nu-1}(z)/z$ reduces it to the displayed identity $A_{\nu,k}(r) > 0$. Division by the negative $R\mathcal{M}_\nu'(z)$ proves that $\alpha_h^\dagger - \alpha < 0$. □

Proof of Propositions 2–3. Retaining both displayed terms of (37) gives

$$d_h = 1 + W_{\nu,k}(h) + \begin{cases} O(h^{2\nu+2}), & 0 < \nu < 1, \\ O(h^4), & 1 < \nu < 2. \end{cases}$$

At $\nu = 1$, the corresponding error is $O\{h^4\log(1/h)\}$. Combining these denominators with (36) and using the bounded derivative of the inverse Matérn map proves (20). For $\nu = 1 + \varepsilon$,

$$b_{1+\varepsilon} = \frac{1}{4\varepsilon} + \frac{2\gamma_{\mathrm{E}} - 1 - 2\log 2}{4} + O(\varepsilon), \tag{40}$$

$$m_{2+2\varepsilon}(\alpha h)^{2+2\varepsilon} = (\alpha h)^2\Big[m_2 + 2\varepsilon\{\ell_{2,k} + m_2\log(\alpha h)\} + O(\varepsilon^2)\Big]. \tag{41}$$

The pole cancels the analytic quadratic term, leaving (18) at $\nu = 1$.

For directions e_1, e_2, the common d_h cancels and

$$\rho_h(Re_1) - \rho_h(Re_2) = h^2\{B_{\nu,k}(Re_1) - B_{\nu,k}(Re_2)\} + o(h^2).$$

The inverse-function theorem and

$$\mathcal{M}_\nu''(z) - \mathcal{M}_\nu'(z)/z = 2^{1-\nu}z^\nu K_{\nu-2}(z)/\Gamma(\nu)$$

then give (21) for every fixed $\nu > 0$. □

B PROOF OF THE FINITE-LIBRARY CERTIFICATE

Proof of Proposition 4. Write $X_i = \Sigma_0^{1/2}Z_i$, stack the independent $Z_i \sim N_p(0, I_p)$, and put $A_\theta = \Sigma_0^{1/2}\Sigma_\theta^{-1}\Sigma_0^{1/2}$. The random likelihood term is the quadratic form associated with $I_N \otimes A_\theta$. The two-sided Gaussian quadratic-form inequality gives, for $t > 0$,

$$\begin{aligned}\Pr\Big(\Big|\sum_{i=1}^N X_i^\top\Sigma_\theta^{-1}X_i - N\operatorname{tr}(A_\theta)\Big| &> 2\sqrt{N}\|A_\theta\|_F\sqrt{t} + 2\|A_\theta\|_{\mathrm{op}}t\Big) \\ &\le 2e^{-t}.\end{aligned} \tag{42}$$

Indeed, the Frobenius and operator norms of the Kronecker matrix are $\sqrt{N}\|A_\theta\|_F$ and $\|A_\theta\|_{\mathrm{op}}$. The log determinant is deterministic. Divide by $2Np$, take $t = \log(2M/\delta)$, and union bound over the deterministic library to obtain (32). On that simultaneous event, the empirical-minimizer comparison through $\widehat{L}_N$ gives

$$L(\widehat{\theta}) - L(\theta^\star) \le [L - \widehat{L}_N](\widehat{\theta}) + [\widehat{L}_N - L](\theta^\star) \le 2\max_\theta u_\theta.$$

Finally, the Gaussian KL formula differs from $pL(\theta)$ only by terms independent of θ. If $c\Sigma_0 \preceq \Sigma_\theta$, congruence after inversion gives $A_\theta \preceq c^{-1}I_p$, hence $\|A_\theta\|_{\mathrm{op}} \le c^{-1}$ and $\|A_\theta\|_F \le \sqrt{p}/c$, which proves (35). □